

the models with their own data.

Netrasemi: Kerala-based semiconductor startup Netrasemi is developing edge AI system-on-chip (SoC) for applications like surveillance, robotics, smart infrastructure, and industrial AI. By integrating dedicated neural processing units or AI accelerators directly on-chip, edge AI SoCs enable real-time AI computing on the edge device itself, eliminating the need to offload inference tasks to external GPUs. Purpose-built edge AI SoCs thereby bring down the latency, cost, and power consumption significantly, compared to traditional GPU-based approaches. The company aims to bring four AI- and video-enabled SoC variants into production by early 2026. It has raised venture capital of ₹1.07 billion from Zoho Corporation and Unicorn India Ventures, with Zoho also partnering for R&D.

Making AI run on normal CPUs

If custom silicon is an emerging trend in the hallowed halls of the hyperscalers, CPU-based AI is emerging as a disruptive force among smaller players. This approach uses advanced optimisation techniques such as quantisation, pruning, and specialised runtimes to enable normal CPUs to run AI workloads. However, it is generally agreed that only inferencing and light training is possible using CPUs, and that too for smaller models. The heavy lifting must be done by GPUs, which may be hired on the cloud.

However, using CPUs instead of GPUs at the edge and for inferencing itself will create a big difference. It brings down costs and improves sustainability. As applications run on-premise, it enhances privacy and data security in sensitive applications like healthcare, banking, etc. It fuels the ubiquity of AI, enabling its deployment on edge devices such as mobile phones, drones, and smart meters.

Enabling AI on standard CPUs generally involves optimising the AI



Innovative solutions, such as Kompact AI from Indian startup Ziroh and IIT Madras, are defying the monopoly of GPUs and enabling AI on common CPUs (Courtesy: Ziroh)

model, making the most of a CPU's special features, and using the right software libraries.

As far as model optimisation goes, there is a focus on building small language models that can achieve specific tasks with fewer computational resources. Another popular technique is quantisation, which reduces the precision of a model's numerical weights (for example, from 32-bit to 8-bit), and consequently the computing power and resources required to process the data. Then, there is binarisation of weights, which limits the parameters of a model to only a few discrete values, such as 0 and 1, or -1, 0, and 1. This drastically reduces resource consumption and memory requirements. Pruning involves eliminating unnecessary connections within a neural network. These connections often do not significantly impact the model's accuracy, and removing them can save computing resources.

It is also important to leverage a CPU's architectural strengths for AI workloads. This involves splitting computations across multiple CPU cores and employing single-instruction, multiple-data instructions to process multiple data pieces simultaneously within each core. Large CPU caches are also leveraged to reduce memory

latency. Choosing the right software frameworks and runtimes is crucial to how AI models interact with CPU hardware. Open source stacks offer a significant long-term advantage.

AI on CPUs—some key projects

Here are some interesting projects enabling edge AI or AI inferencing using commodity hardware.

Microsoft BitNet: Motivated by the need to run LLMs efficiently on common hardware like CPUs and edge devices, Microsoft developed Bitnet in association with industry and academic research partners. Bitnet uses a quantisation scheme that restricts all weights to ternary values -1, 0, and 1, eliminating floating-point operations. This significantly reduces memory and energy requirements, while reasonably matching the performance of comparable GPU-based open-weight models. This makes it possible to run LLMs on commodity hardware. Bitnet comprises a model family (like BitNet b1.58 2B4T, a 2-billion-parameter LLM trained on 4 trillion tokens) and an open source inference framework (bitnet.cpp) optimised for these 1-bit models. According to reports, BitNet b1.58 2B4T achieves performance on par with top open-weight models of similar